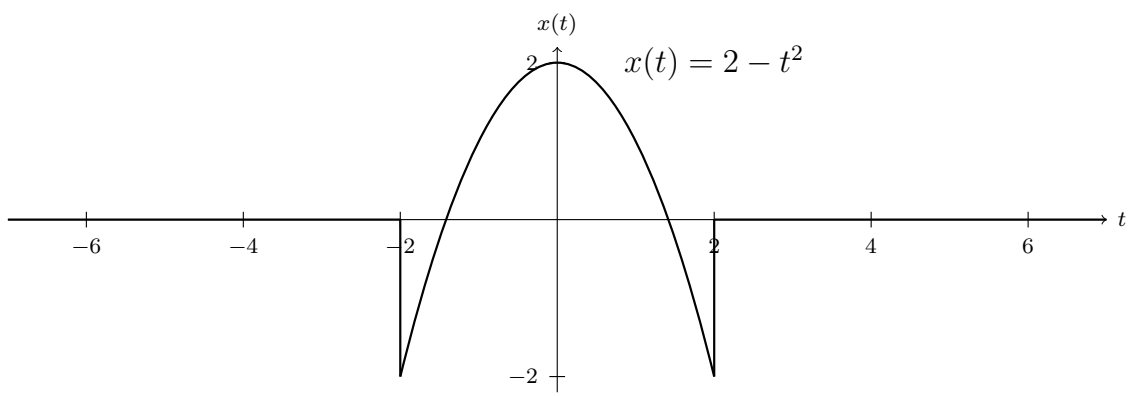
ECE 3710 – Quiz 2 – Energy and Power

Name: Solution Work on Last Page →

T#: _____



Based on the figure above, choose the ***best*** answer for Questions 1-3. Then, solve Question 4 using the provided power equations. Show your work and circle your answers.

1. (1 point) The signal $x(t)$ is periodic or $\boxed{\text{aperiodic}}$ **(circle one).**
2. (1 point) The signal $x(t)$ is an $\boxed{\text{energy signal,}}$ a power signal, or neither **(circle one).**
3. (6 points) If the signal $x(t)$ is an $\boxed{\text{energy signal,}}$ then compute its energy. If the signal $x(t)$ is a power signal, then compute its power.

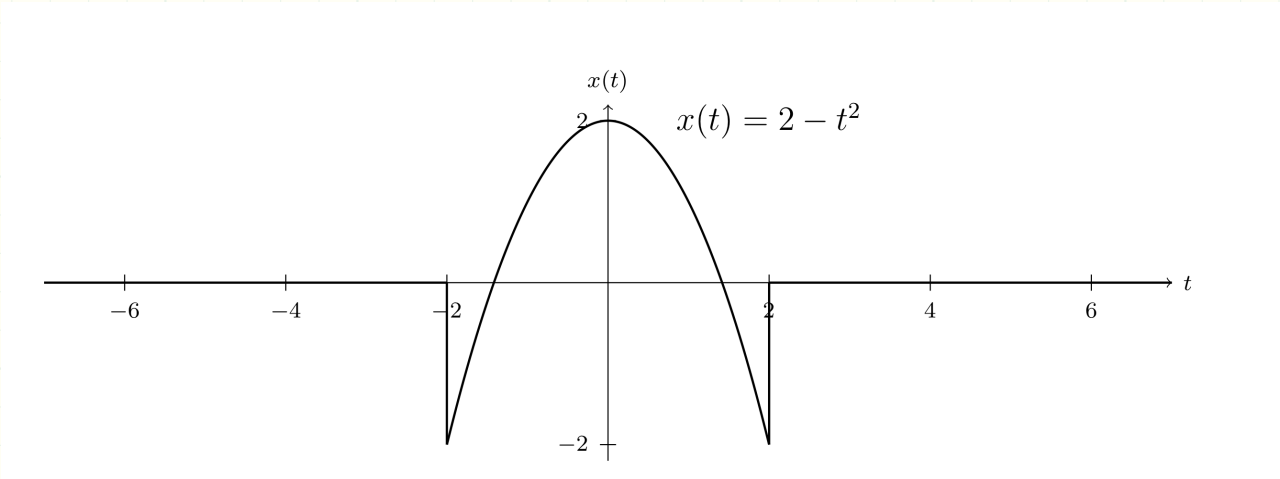
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt \qquad P_x = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |x(t)|^2 dt \underbrace{= \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt}_{\text{For periodic signals.}}$$

$$\boxed{E_x = 7.467 \text{ J}}$$

4. (2 points) Calculate the logarithmic power gain (in dB) of $x_3(t)$ relative to $x_2(t)$.

$$G_p(\text{dB}) = 10\log_{10}\left(\frac{P_3}{P_2}\right), \qquad x_2(t) = \cos(\omega t + \theta_2), \qquad x_3(t) = 4\cos(\omega t + \theta_3)$$

$$\boxed{G_p(\text{dB}) = 12 \text{ dB}}$$



Problem ①:

$\boxed{\text{The signal } x(t) \text{ is aperiodic.}}$

Problem ②:

To determine if $x(t)$ is an energy signal, power signal, or neither, we first check if it has finite energy!

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-2}^{2} |2-t^2|^2 dt = \int_{-2}^{2} (2-t^2)(2-t^2) dt$$

$$E_x = \int_{-2}^{2} (4 - 4t^2 + t^4) dt = \left(4t - \frac{4}{3}t^3 + \frac{1}{5}t^5\right)\Big|_{-2}^{2}$$

$$E_x = \left(8 - \frac{32}{3} + \frac{32}{5}\right) - \left(-8 + \frac{32}{3} - \frac{32}{5}\right) = 7.467 \text{ Joules}$$

$\boxed{\text{Since } E_x \text{ has finite energy, } x(t) \text{ is an Energy Signal.}}$

Problem ③:

$\boxed{E_x = 7.467 \text{ Joules}}$

Problem ④: Given Power formula → But Given waveform

$$G_p(dB) = 10 \log_{10}\left(\frac{P_3}{P_2}\right) \; ; \quad P_2 = \frac{1^2}{2} = \frac{1}{2} \qquad P_3 = \frac{4^2}{2} = 8$$

$\boxed{G_p(dB) = 12 \text{ dB}}$

$$10 \log_{10}\left(\frac{8}{\frac{1}{2}}\right) = 10 \log_{10}(16) = 12 \text{ dB}$$